40) Let $f(x)=ax+b$ and $g(x)=cx+d$, where a,b,c and d are constants. Determine necessary and sufficient conditions on the constants a,b,c and d so that $fog=gof$.

Solution: $f(g(x))=f(cx+d)=a(cx+d)+b=acx+ad+b$

$g(f(x))=g(ax+b)=c(ax+b)+d=acx+bc+d$.

Let, $fog=gof \Rightarrow acx+ad+b=acx+bc+d \Rightarrow \boxed{d(a-1)=b(c-1)}$

$\therefore d(a-1)=b(c-1)$ is a necessary condition for $fog=gof$.

It's also easy to show that its a sufficient condition. □

41) Show that the function $f(x)=ax+b$ from R to R, where a and b are constants with $a\neq 0$ is invertible, and find the inverse of f.

Solution: Let, $x,y\in R$ and $f(x)=f(y)$. $\therefore ax+b=ay+b \Rightarrow ax=ay \Rightarrow x=y$. By Defn. 5, f is one-to-one.

Let, $f(x)=Z$ where $Z\in R$. $\therefore ax+b=Z \Rightarrow \boxed{x=\frac{Z-b}{a}}$. $\because a\neq 0$, its easy to see that $x\in R$. By Defn. 7, $\because \forall Z\in R, \exists x\in R$ s.t. $f(x)=Z$, $\therefore f$ is onto.

By Defn. 8, f is a bijection. By Defn. 9, f is invertible.

$f(x)=Z \Rightarrow f^{-1}(Z)=x \Rightarrow \boxed{f^{-1}(Z)=\frac{Z-b}{a}}$ □

42) Let f be a function from the set A to the set B. Let S and T be subsets of A. Show that

a) $f(S\cup T)=f(S)\cup f(T)$.

Solution: Given, $f: A\to B$. Given, $S\subseteq A$ and $T\subseteq A$.

By Defn. 3 from Sec-2.1, $\forall x, x\in S \to x\in A$.

By Defn. 3 from Sec-2.1, $\forall x, x\in T \to x\in A$.

$S\cup T=\{x \mid x\in S \vee x\in T\}$ (By Defn. 1 from Sec-2.2)

$f(S\cup T)=\{f(x) \mid f(x)\in f(S) \vee f(x)\in f(T)\}$

$= f(S)\cup f(T)$ (By Defn. 1 from Sec-2.2)